

Project Design Phase

Solution Architecture

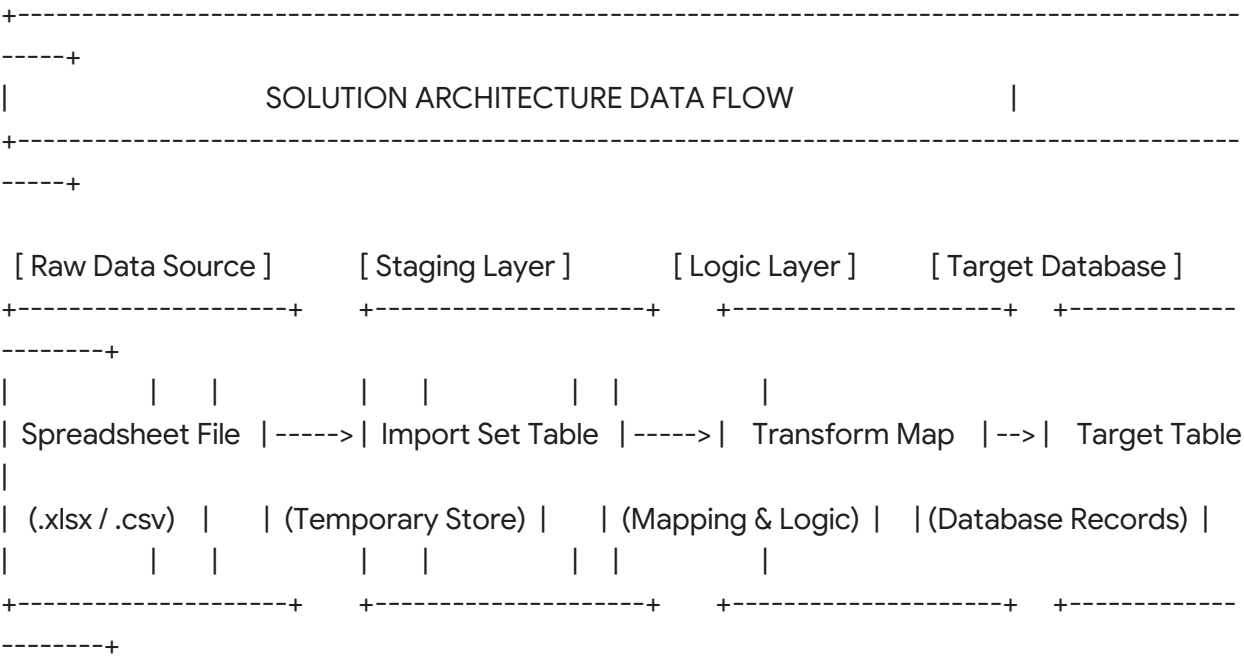
Project Name: Import Data Using Transform Maps (Spreadsheet)

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- **Find the best tech solution to solve existing business problems:** Automate and streamline bulk data ingestion by uploading external spreadsheet records (Excel/CSV) directly into target database tables, eliminating human error and manual data entry.
- **Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders:** Establish a clear data ingestion pipeline starting from raw file upload through staging tables, transformation rules, and final target record population.
- **Define features, development phases, and solution requirements:** Configure field-to-field mapping rules, identify Coalesce unique identifiers to prevent record duplication, and set up transformation logic for data consistency.
- **Provide specifications according to which the solution is defined, managed, and delivered:** Detail logging mechanisms, staging table cleanup, error handling, and security role access required for execution.

Solution Architecture Diagram:



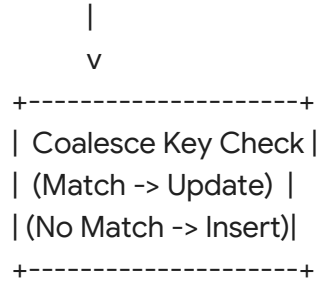


Figure 1: Architecture and data flow of the Import Data Using Transform Maps (Spreadsheet) process

Process Workflow & Technical Specification:

1. Data Source Setup:

- User uploads the external Excel or CSV file containing raw records.

2. Import Set Staging:

- The platform automatically builds an **Import Set Table** (Staging Table) mirroring the structure of the uploaded file columns.

3. Transform Map Configuration:

- Source fields are mapped to destination fields on the target table using **Auto-Map Matching** or **Mapping Assist**.
- **Coalesce Configuration:** A specific field (e.g., *User ID*, *Email*, *Asset Code*) is set to Coalesce = True to enforce deduplication (updating existing records if a match is found, or inserting new ones if no match exists).

4. Target Table Execution:

- The transform engine processes each row from the staging table, executes business rules/scripts, and commits the cleansed records into the destination table.